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# Gut dysbiosis in multiple sclerosis patients: a comparative analysis in fecal samples

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**Multiple sclerosis (MS) is a chronic, inflammatory disease of the central nervous system characterized by neuroinflammation, demyelination, and axonal damage. Recent evidence suggests that gut microbiota may play a critical role in MS pathogenesis by modulating immune responses and contributing to systemic inflammation. Alterations in the abundance of specific bacterial species, such as *Bacteroides fragilis*, *Faecalibacterium prausnitzii*, and *Bifidobacterium spp.*, have been linked to immune dysregulation in MS. However, the role of gut microbiota in treatment-naïve patients at the earliest stages of MS remains underexplored. This case-control study, conducted from 2021 to 2023, included 36 treatment-naïve patients with MS (PwMS) who had experienced their first attack of symptoms and fulfilled McDonald's criteria, and 36 age- and sex-matched healthy controls. Stool samples were collected and analyzed using quantitative PCR (qPCR) to determine the relative abundance of *Bacteroides fragilis*, *Faecalibacterium prausnitzii*, and *Bifidobacterium spp.*. Statistical analyses compared bacterial abundance between groups while controlling for key demographic variables. The MS group included 9 males (25%) and 27 females (75%), with a mean age of 38.3 ± 9.5 years, compared to 39.8 ± 10.2 years in controls. PwMS exhibited significantly lower levels of *Bacteroides fragilis* ($0.11 \times 10^9$ vs. $0.22 \times 10^9$ CFU, $p < 0.05$) and *Bifidobacterium spp.* ($1.50 \times 10^9$ vs. $1.65 \times 10^9$ CFU, $p < 0.05$) compared to controls. Although *Faecalibacterium prausnitzii* levels were lower in PwMS ($1.08 \times 10^9$ vs. $1.27 \times 10^9$ CFU), the difference was not statistically significant. These findings align with emerging evidence suggesting a potential role of gut dysbiosis in MS pathogenesis. The reduced abundance of beneficial bacteria, particularly *B. fragilis* and *Bifidobacterium spp.*, may contribute to altered immune regulation and increased inflammation in PwMS. While these results contribute to our understanding of the gut-brain axis in MS and may inform future therapeutic approaches, further research, including larger longitudinal studies, is needed to clarify the complex relationship between gut microbiota and MS development and progression.**

**Keywords** Gut microbiota, Multiple sclerosis, Bacteroides fragilis, Faecalibacterium prausnitzii, Bifidobacterium spp

Multiple sclerosis (MS) is a chronic, inflammatory, demyelinating disease of the central nervous system (CNS) that affects approximately 2.8 million people worldwide[1]. It is characterized by the destruction of myelin sheaths surrounding nerve fibers, leading to a wide range of neurological symptoms and disabilities[2]. MS typically onsets in young adults, with a mean age of diagnosis around 32 years, and is more prevalent in women than men[3]. The disease can cause various complications, including visual impairment, mobility issues, cognitive dysfunction, and fatigue, significantly impacting patients' quality of life[4]. Although MS is rarely directly fatal, it can reduce life expectancy by 5–10 years and increase the risk of mortality from complications[5]. The economic burden of MS is

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substantial, with annual costs per patient in the United States estimated at $85.4 billion in 2022, including direct medical costs and indirect costs due to lost productivity[6].

MS often begins with a first clinical attack of neurological symptoms, which is referred to as Clinically Isolated Syndrome (CIS). CIS is defined as a monophasic clinical episode with symptoms and objective findings indicative of inflammatory demyelination, such as optic neuritis, partial transverse myelitis, or brainstem syndromes. These episodes last at least 24 h, occur in the absence of fever or infection, and resemble typical MS relapses. The non-life-threatening symptoms such as sensory disturbances, vertigo, or mild motor deficits, frequently prompt patients to visit neurology clinics for evaluation. Diagnosis of MS at this stage often relies on magnetic resonance imaging (MRI) findings demonstrating dissemination in space and/or time according to the McDonald Criteria. While not all patients with CIS progress to MS, many do, particularly when imaging reveals characteristic inflammatory lesions in the brain or spinal cord[7,8]. Studying patients at this early stage provides critical insights into the disease before significant progression or treatment[7].

Recent research has highlighted the potential role of the gut microbiome in the pathogenesis and progression of MS[8]. Specifically, alterations in the abundance of certain bacterial species, such as *Bacteroides fragilis*, *Faecalibacterium prausnitzii*, and *Bifidobacterium spp.*, have been associated with immune system modulation and neuroinflammation[9,10]. While much of the existing literature focuses on established MS patients, the gut microbiota of treatment-naïve patients who have experienced their first attack remains largely unexplored. This population provides an ideal opportunity for studying early disease mechanisms without the confounding effects of immunomodulatory therapies.

The gut microbiome has emerged as a critical factor in human health and disease, with growing evidence suggesting its involvement in neurological disorders, including MS. Several key studies have investigated the relationship between gut microbiota and MS reported alterations in the gut microbiome of MS patients, with increased abundance of Methanobrevibacter and Akkermansia and decreased levels of Butyricimonas[11]. These changes were associated with expressions of genes involved in dendritic cell maturation, interferon signaling, and NF-κB signaling pathways. Cekanaviciute et al. demonstrated that gut microbiota from MS patients could exacerbate experimental autoimmune encephalomyelitis (EAE) in mice, suggesting a causal role of gut bacteria in MS pathogenesis[12]. They also observed reduced levels of Faecalibacterium and increased Akkermansia in MS patients. A systematic review synthesized findings from multiple studies, confirming consistent alterations in gut microbiota composition in MS patients compared to healthy controls[13]. However, the review also highlighted the variability in results across studies, emphasizing the need for further research. Despite growing evidence linking gut microbiota and MS, the specific roles of key bacterial species, including Bacteroides fragilis, Faecalibacterium prausnitzii, and *Bifidobacterium spp*., in treatment-naïve patients following their first clinical attack remain unclear[14]. These bacteria are implicated in immune regulation, gut barrier function, and neuroinflammation but have not been consistently characterized in this early stage of the disease. Understanding their relative abundance in this population compared to healthy controls could help elucidate the mechanisms by which gut microbiota influence MS pathogenesis and progression[15]. Our selection of *Bacteroides fragilis*, *Faecalibacterium prausnitzii*, and Bifidobacterium spp. was based on their well-established immunoregulatory functions rather than prevalence in metagenomic surveys. *Bacteroides fragilis* produces polysaccharide A that promotes regulatory T cell responses; F. prausnitzii is a major butyrate producer involved in gut anti-inflammatory signaling; and *Bifidobacterium spp.* are associated with gut barrier protection and metabolic immune interactions. Thus, these taxa represent mechanistically relevant targets in the context of early MS pathogenesis[14].

In particular, the International Multiple Sclerosis Microbiome Study (iMSMS) consortium has provided one of the most comprehensive and standardized analyses of the gut microbiome in MS patients to date, identifying reproducible microbial signatures associated with the disease across multiple populations. This work further highlights the importance of functional microbial traits in MS pathophysiology and supports the relevance of mechanistically informed species selection in microbiota studies[16].

The primary objective of this study is to compare the relative abundance of *Bacteroides fragilis*, *Faecalibacterium prausnitzii*, and *Bifidobacterium spp*. in fecal samples of treatment-naïve PwMS and healthy controls.By focusing on this early disease stage, the study aims to provide insights into the gut-brain axis and its potential role in neuroinflammatory diseases.

## Methods and materials

### Study population

This case-control study was conducted in Tehran, Iran, between May 2021 and March 2023 to compare analysis of *Bacteroides fragilis*, *Faecalibacterium prausnitzii*, *Bifidobacterium spp.* in fecal samples of PwMS who had experienced their first attack of symptoms and healthy controls. The study protocol received approval from the Ethics Committee at Tehran Azad University of Medical Sciences (IR.IUMS.TMU.REC.1400.015) and adhered to the Declaration of Helsinki principles[17].

In total, 72 individuals, comprising 36 PwMS and 36 healthy controls, were enrolled. PwMS were recruited from neurology clinics following their first clinical attack and were diagnosed using McDonald's criteria based on clinical evaluations, MRI, and cerebrospinal fluid analysis[18]. PwMS had to meet the following inclusion criteria:

- Diagnosed with MS based on the McDonald Criteria following their first clinical attack of neurological symptoms.
- Neurological symptoms included non-life-threatening sensory disturbances, vertigo, or mild motor deficits.
- No history of previous clinical episodes of demyelination.
- Treatment-naïve, with no prior use of disease-modifying therapies (DMTs), corticosteroids, or immunosuppressive drugs.
- Clinically stable, without acute infections or inflammatory conditions in the past six months.

- No antibiotic use in the past three months.
- No gastrointestinal disorders such as inflammatory bowel disease (IBD) or irritable bowel syndrome (IBS).
- Body Mass Index (BMI) under 30 (classified as non-obese).
- No psychoactive medication use or abdominal surgeries in the past six months.

The control group comprised 36 age- and sex-matched healthy controls, selected from hospital staff and their immediate social networks. Their inclusion criteria were as follows: No personal or family history of autoimmune diseases, including MS, Free from infectious or inflammatory diseases for at least six months, No antibiotic use for three months prior to stool sample collection, No history of gastrointestinal disorders, such as IBD or IBS, No corticosteroid use in the past six months, No chemotherapy treatment in the past year, No previous surgeries involving the gastrointestinal tract, No use of psychoactive medications in the past six months, No diagnosis of colorectal cancer, metabolic syndrome, or diabetes, BMI under 30 (non-obese).

Participants were excluded from the study if they met any of the following criteria: Declined to participate or withdrew consent, Recent infectious or inflammatory conditions (within the past six months), Use of antibiotics, immunosuppressive therapies, or psychoactive medications in the past three months, Diagnosis of gastrointestinal diseases like IBD or IBS, History of abdominal surgery within the last six months, Diagnosis of colorectal cancer, metabolic syndrome, or diabetes, BMI above 30 (classified as obese), Presence of malnutrition or other metabolic disorders.

Stool samples were collected from all participants during a clinically stable phase, with no recent antibiotic use or other confounding medical interventions. Informed consent was obtained from all subjects and/or their legal guardian(s). All patients were included shortly after diagnosis and following their first demyelinating episode, minimizing the potential confounding effects of long-term disease progression or treatment.

## Sample size

The sample size for this case-control study was calculated to detect a significant difference in the relative abundance of key gut microbiota species between treatment-naïve PwMS and healthy controls. The calculation was based on the following assumptions:

Type I error (α): 0.05 (5% significance level).

Type II error (β): 0.20 (80% power).

Effect size (d): Based on previous studies comparing gut microbiota in MS patients and healthy controls, we assumed a moderate effect size of 0.5 for the difference in bacterial abundance.

Standard deviation (σ): Estimated from the variability of bacterial counts reported in prior studies.

Two-tailed test.

The sample size was calculated using the following formula for comparing two independent means:

$$n = \frac{(Z_{\alpha/2} + Z_{\beta})^2 \times 2 \times \sigma^2}{\Delta^2}$$

$Z_{\alpha/2}$= Z-score for the desired significance level.

$Z_{\beta}$= Z-score for the desired power.

σ = Standard deviation of the outcome variable (bacterial abundance).

Δ= Minimum detectable difference between the two groups (effect size)

$$n = \frac{(1.96 + 0.84)^2 \times 2 \times (0.05)^2}{(0.10)^2} = 33$$

With considering 10% attrition rate in each group, the calculations indicated a need for 36 participants in each group to adequately address the research objectives.

## Gut microbiota assessment

*Sample collection and handling*

Fresh stool samples were collected from participants and immediately transported under cold-chain conditions (maintained at 4 °C) to the Pasteur Institute of Iran for further processing. To preserve bacterial viability, stool samples were placed in sealed plastic bags containing oxygen absorbers and $CO_2$ producers to maintain an anaerobic environment. The samples were then suspended in phosphate-buffered saline (PBS) solution containing 20% glycerol and rapidly frozen in liquid nitrogen. All samples were stored at −80 °C until DNA extraction and analysis.

*DNA extraction from fecal samples*

Genomic DNA was extracted from stool samples using the FavorPrep™ Stool DNA Isolation Mini Kit (Favorgen Biotech, Taiwan) according to the manufacturer's protocol. This kit allows for rapid and efficient purification of high-quality DNA from both fresh and frozen fecal material, making the isolated DNA suitable for downstream applications such as real-time PCR, enzymatic reactions, or long-term storage at − 20 °C. Upon removal from the freezer, all samples were thawed and maintained on ice throughout the extraction procedure to preserve DNA integrity.

The extraction process included the following sequential steps: sample preparation, centrifugation, cell lysis, DNA binding, washing, and elution. Briefly, approximately 200 mg of stool was transferred to a bead-containing lysis tube and placed on ice. Subsequently, 300 µL of SED1 buffer and 20 µL of Proteinase K were added. The mixture was vortexed at maximum speed for 5 min and incubated at 60 °C for 20 min, with vortexing every

5 min to ensure complete homogenization. After incubation, the sample was cooled, and 100 µL of SDE2 buffer was added. The tube was vortexed and incubated on ice for 5 min.

Following centrifugation at 14,000 rpm for 5 min, the supernatant was carefully transferred to a 1.5 mL microcentrifuge tube, avoiding the pellet. To the collected supernatant, 200 µL of SDE3 buffer was added, mixed, and incubated at room temperature for 2 min. After a second centrifugation (2 min, 14,000 rpm), 250 µL of the supernatant was transferred to a new tube, followed by the addition of 250 µL of SDE4 buffer and 250 µL of 96–100% ethanol. The mixture was vortexed thoroughly and applied to an SDE spin column.

The column was centrifuged for 1 min, and the flow-through was discarded. It was then washed twice with 750 µL of ethanol-based wash buffer. To dry the membrane, the column was centrifuged at full speed for 3 min. Finally, DNA was eluted with 200 µL of preheated elution buffer ($ddH_2O$), added to the center of the membrane, incubated at room temperature for 2 min, and centrifuged for 1 min. All centrifugation steps were performed at room temperature (15–25 °C) and 14,000 rpm (≈ 20,000 × g). The extracted DNA was stored at − 80 °C until further use. DNA concentration was measured at 260 nm prior to storage.

*DNA quantification by nanodrop*

The purity and concentration of the extracted DNA were assessed using a NanoDrop spectrophotometer (Thermo Scientific Nano), which measures absorbance at 260 nm for nucleic acids and at 280 nm for protein contamination. The device was initially blanked using 2 µL of the elution buffer (ATE), and then 2 µL of each DNA sample was loaded using a crystal pipette tip. The A260/A280 ratios for all samples ranged between 1.8 and 2.0, confirming adequate DNA purity for molecular analysis.

*Quantitative real-time PCR analysis of gut microbiota*

To assess the relative abundance of Bacteroides fragilis, Faecalibacterium prausnitzii, and Bifidobacterium spp., species-specific primers targeting the 16 S rRNA gene were employed. Where necessary, primers were designed using the NCBI Primer-BLAST tool, based on the nucleotide sequences of the target bacterial species. To ensure specificity and eliminate potential cross-reactivity or sequence homology with non-target organisms, all primer sequences were validated using the NCBI BLAST search tool (https://pmlegacy.ncbi.nlm.nih.gov/tools/primer-blast). The complete list of primer sequences utilized in this study is provided in the table below.

Quantitative PCR (qPCR) reactions were performed using a Rotor-Gene Q thermal cycler (Qiagen, Germany), equipped with a 36-tube rotor. All reagents, including the extracted DNA, specific primers, and SYBR® master mix, were kept on ice during assay preparation to preserve reagent stability. Each reaction had a total volume of 10 µL, consisting of 5 µL of SYBR® Premix Ex Taq II (Takara, Japan), 0.5 µL of each primer (forward and reverse), 1 µL of template DNA, and 3 µL of nuclease-free water. The qPCR thermal profile included an initial denaturation step at 94 °C for 4 min, followed by 40 amplification cycles comprising denaturation at 95 °C for 30 s, annealing at 58 °C for 45 s, and extension at 72 °C for 45 s. All reactions were run in duplicates. Negative controls (no template DNA) and a standard curve using serial dilutions of E. coli genomic DNA were included. Although serial dilutions of E. coli genomic DNA were used for the generation of a standard curve to assess PCR amplification efficiency, the final quantification of each bacterial species (*Bacteroides fragilis*, *Faecalibacterium prausnitzii*, and *Bifidobacterium spp*.) was based on species-specific calibration. For each target bacterium, CFU equivalents were estimated by correcting for genome size and the known copy number of the 16 S rRNA gene per genome. DNA extraction efficiency was standardized across all samples by using consistent mass input and lysis conditions. Importantly, while qPCR quantification reflects total DNA rather than viability, CFU equivalents were used for comparative purposes, following established practice in microbiome qPCR studies to express relative abundance. This conversion allows for better interpretability in the context of gut microbial load, though we acknowledge that this does not equate to viable colony counts from culture-based methods[22,23]. QPCR output was analyzed based on amplification curves, and DNA concentrations were converted to CFU using the following formula:

$$\mathrm{CFU} = \frac{(\text{amount (ng)} \times 6.022 \times 10^{23})}{(\text{genome length (bp)} \times 650 \times 10^{9})}$$

Where:
Amount: DNA concentration (ng).
$6.022 \times 10^{23}$: Avogadro's number (molecules/mol).
650: average molecular weight (g/mol) of one base pair.
Genome length: total base pairs in the target bacterial genome.

No bacterial culture was performed in this study. Instead, colony-forming unit (CFU) equivalents were inferred from qPCR data using a molecular calculation that incorporates Avogadro's number, genome length of each species, and average DNA mass per base pair. This method assumes that each genome copy corresponds to a single bacterial cell and was adjusted by species-specific 16 S rRNA gene copy number. Although this approach does not reflect bacterial viability, it is widely employed in microbiome quantification studies for estimating relative abundance in fecal samples.

## Statistical analysis

All statistical analyses were performed using SPSS version 26.0 (IBM Corp, Armonk, NY) and GraphPad Prism 9.0 (GraphPad Software, San Diego, CA). Data distribution was assessed using the Shapiro-Wilk test. For bacterial abundance data that followed a normal distribution, arithmetic means and standard deviations (SD) were reported. In cases where the data were non-normally distributed, medians and interquartile ranges (IQR)

| Target bacteria | Primer sequence (5'→3') | Annealing temp (°C) |
|---|---|---|
| Bacteroides fragilis[19] | F: CGATGGATAGGGGTTCTGAGAG<br>R: GCTGATAGGCGTCCGTCTT | 60 |
| Faecalibacterium prausnitzii[20] | F: GATGGCCTCGCGTCCGATTAG<br>R: CCGAAGACCTTCTTCCTCC | 58 |
| Bifidobacterium spp[21]. | F: TCGCGTCYGGTGTGAAAG<br>R: CCCCACATCCAGCRTCCAC | 60 |

**Table 1**. Primer sequences and PCR conditions for target bacteria.

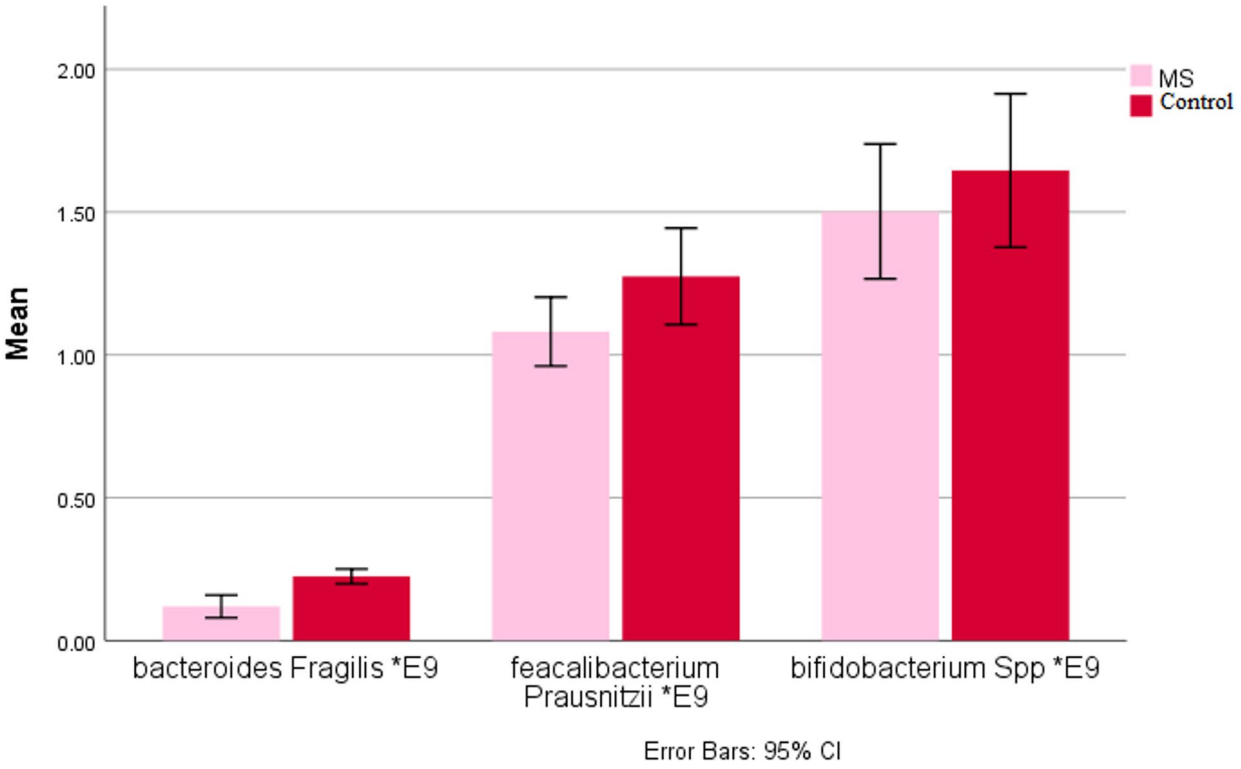



**Fig. 1**. Frequency of Bacteroides Fragilis, Feacalibacterium Prausnitzii and bifidobacterium spp in both control and MS groups. The numbers reported are on a CFU basis.

were used instead. This distinction was applied throughout the analysis to account for the known skewness in microbiota-related data.

Categorical variables were expressed as frequencies and percentages. Independent samples t-tests were used to compare normally distributed bacterial abundance data between the treatment-naïve PwMS and control groups. For non-normally distributed data, the Mann-Whitney U test was applied. The statistical significance threshold was set at $p < 0.05$ for all analyses.

## Results

The study participants included 36 treatment-naïve PwMS, consisting of 9 males (25%) and 27 females (75%). The average age of PwMS was 38.3 years, while the healthy control had an average age of 39.8 years. MS symptoms typically began around age 34.8 for the patients. Nearly half (44%) of the PwMS group were current smokers. A summary of the participants' baseline characteristics demonstrated in Table 1.

The study compared the levels of the mentioned gut bacteria in two groups by counting colony-forming units (CFU). The control group showed higher average counts than MS patient group for three key bacteria:

*Bacteroides fragilis*: $0.22 \times 10^9$ in controls vs. $0.11 \times 10^9$ in MS patients *Faecalibacterium prausnitzii*: $1.27 \times 10^9$ in controls vs. $1.08 \times 10^9$ in MS patients.

*Bifidobacterium spp.*: $1.65 \times 10^9$ in controls vs. $1.50 \times 10^9$ in MS patients (Fig. 1**).** We found statistically significant differences ($P < 0.05$) between the control and MS groups for *Bacteroides fragilis* and *Bifidobacterium spp*. (Table 2). Table 3 shows that treatment-naïve MS patients had significantly lower levels of Bacteroides fragilis ($p < 0.001$) and Bifidobacterium spp. ($p = 0.04$) compared to healthy controls. Faecalibacterium prausnitzii also showed a decreasing trend in the MS group, but the difference was not statistically significant ($p = 0.06$). These results suggest early gut dysbiosis in MS, particularly involving key anti-inflammatory bacterial species.

| | MS ($n=36$) | Control ($n=36$) | $P$-value[a] |
|---|---|---|---|
| Sex | | | |
| Male | 9 (25%) | 9 (%) | 0.41 |
| Female | 27(75%) | 27(75%) | 0.56 |
| Age (year) | 38.3 ± 9.5 | 39.8 ± 10.2 | 0.92 |
| Age of disease onset | 34.8 ± 3.5 | - | - |
| BMI (kg/m$^2$) | 27.8 ± 4.1 | 26.5 ± 5.4 | 0.23 |
| Married% | 19 (46%) | 22(53%) | 0.12 |
| University graduate% | 17(39%) | 26(60%) | 0.04 |
| Smoke% | 20 (44%) | 25(55%) | 0.06 |

**Table 2**. General characteristics of the study participants based on cases and controls. ∗Obtained from independent-samples $t$ test or $\chi^2$ test, where appropriate. MS: Multiple Sclerosis.

| Gut Microbiota | Control (CFU) ($n=36$) | MS (CFU) ($n=36$) | *$P$-value |
|---|---|---|---|
| Bacteroides fragilis | $0.22\times10^9 \pm 0.07$ | $0.11\times10^9 \pm 0.11$ | < 0.001 |
| Faecalibacterium prausnitzii | $1.27\times10^9 \pm 0.5$ | $1.08\times10^9 \pm 0.3$ | 0.06 |
| *Bifidobacterium spp*. | $1.65\times10^9 \pm 0.8$ | $1.50\times10^9 \pm 0.7$ | 0.04 |

**Table 3**. Mean and standard deviation of bacterial strains in MS and control groups. *Obtained from independent T-test (adjusted for, Age, Sex, and university graduate). MS: Multiple Sclerosis.

## Discussion

This study demonstrates significant gut microbiota alterations in treatment-naïve PwMS following their first clinical attack. Reduced levels of *Bacteroides fragilis* and *Bifidobacterium spp.* were observed in PwMS compared to healthy controls, supporting the hypothesis that gut dysbiosis may contribute to early immune dysregulation and neuroinflammation in MS. Although *Faecalibacterium prausnitzii* levels were lower in PwMS, the difference was not statistically significant.

Our findings are consistent with previous studies exploring the relationship between gut microbiota and MS. Jangi et al. (2016) reported significant alterations in the gut microbiome of MS patients, including reduced levels of *Bacteroides* species, which may be associated with immune dysregulation and neuroinflammation[11]. Similarly, Tremlett et al. (2017) found a reduction in *Bacteroides* species in pediatric MS patients, further supporting the hypothesis that these bacteria may play a protective role through their anti-inflammatory properties[24]. The observed reduction in *Bacteroides fragilis* in our study is particularly noteworthy, as this species is known to produce polysaccharide A, a molecule that induces regulatory T cells and suppresses neuroinflammation. This aligns with animal studies, such as those by Ochoa-Repáraz et al. (2010), which demonstrated that *Bacteroides fragilis* could protect against(CNS demyelination in mouse models of MS.

Regarding Faecalibacterium prausnitzii, our study found a non-significant reduction in MS patients compared to controls. This result contrasts with the findings of Cekanaviciute et al. (2017), who reported a significant decrease in *Faecalibacterium prausnitzii* in MS patients, which was associated with increased disease severity[12]. *Faecalibacterium prausnitzii* is a known producer of butyrate, a short-chain fatty acid (SCFA) with anti-inflammatory properties, and its depletion has been linked to impaired gut barrier function and systemic inflammation in various autoimmune disorders[15]. The lack of statistical significance in our findings could be due to differences in study design, sample size, or geographic and dietary factors affecting microbiota composition. Further research with larger cohorts is needed to clarify the role of *Faecalibacterium prausnitzii* in MS.

Our study also observed a significant reduction in *Bifidobacterium spp.* in MS patients, which is consistent with the findings of Cosorich et al. (2017), who reported decreased levels of *Bifidobacterium* in relapsing-remitting MS patients[25]. However, it is important to note that most of the referenced studies used 16 S rRNA sequencing, which generally offers taxonomic resolution at the genus level. In contrast, our study employed species-specific primers in a qPCR platform, enabling direct quantification of *Bacteroides fragilis*, *Faecalibacterium prausnitzii*, and *Bifidobacterium spp*. at the species level. This methodological distinction provides greater specificity and functional relevance when interpreting the differences observed.

Bifidobacterium species are known for their role in maintaining intestinal barrier integrity and producing SCFAs that regulate immune responses. Their reduction in MS patients could contribute to a "leaky gut," allowing the translocation of bacterial products into the bloodstream, potentially triggering systemic inflammation and exacerbating MS symptoms[26]. Additionally, Bifidobacterium has been shown to modulate the gut-brain axis, which plays a critical role in regulating neuroinflammatory processes[27]. Although the present study identified a significant reduction in *Bifidobacterium spp*. in treatment-naïve PwMS, it is important to acknowledge that some previous studies have reported an increased abundance of *Bifidobacterium* in MS patients. For example, some metagenomic studies have detected elevated levels of certain *Bifidobacterium* species in MS, potentially reflecting disease stage, treatment status, or geographic and dietary differences across cohorts[9,24]. These contrasting findings

underscore the complexity of host-microbiota interactions and highlight the need for stratified and longitudinal analyses to better understand the context-dependent role of *Bifidobacterium* in MS pathophysiology.

The differences observed between our study and some previous reports may stem from varying methodologies, including differences in microbiome sampling, DNA extraction protocols, and the use of quantitative PCR (qPCR) versus next-generation sequencing. Moreover, geographic and dietary differences between study populations may also contribute to the variability in gut microbiota composition across studies. While our study provides valuable insights into the gut microbiota profile of MS patients, the cross-sectional design limits our ability to establish causality. Future longitudinal studies and mechanistic research are needed to determine whether these microbial changes are a cause or consequence of MS and to explore the potential for microbiota-based therapeutic interventions.

## Strengths and limitations

The study offers several strengths that enhance the validity of its findings. First, the case-control design allowed for a direct comparison of gut microbiota between MS patients and healthy controls, providing insights into potential microbial dysbiosis associated with MS. Additionally, the study employed quantitative PCR (qPCR) for bacterial quantification, a highly sensitive and specific method for detecting bacterial DNA, which ensured precise measurements of the relative abundance of key bacterial species. The inclusion of strict exclusion criteria, such as no recent antibiotic use or disease-modifying therapies, helped minimize confounding factors that could influence gut microbiota composition. Furthermore, the focus on specific bacterial species with known immunoregulatory roles, like *Bacteroides fragilis* and *Bifidobacterium spp*., adds clinical relevance to the investigation.

However, the study also has notable limitations. The relatively small sample size may limit the generalizability of the results, as gut microbiota composition can vary widely based on individual factors such as diet, lifestyle, and geographic location. The cross-sectional nature of the study precludes establishing causality—whether the observed microbial alterations contribute to MS development or are a consequence of the disease remains unclear. Another limitation is the lack of comprehensive microbiota profiling, which might have captured a broader spectrum of microbial changes that could be relevant to MS pathogenesis. Additionally, we did not measure the Expanded Disability Status Scale (EDSS) in MS patients, which is a key factor for assessing disease severity and its potential correlation with gut microbiota composition. Finally, unmeasured variables such as diet, stress levels, and environmental exposures might still have influenced microbiome composition. Future studies with larger, more diverse populations and longitudinal designs are necessary to confirm these findings and explore causality. Furthermore, we did not measure EDSS scores, which limits our ability to correlate microbiota profiles with disease severity and the absence of total bacterial load quantification using universal 16 S primers. As a result, the bacterial abundance reported for each target species reflects absolute values rather than relative abundance normalized to total bacterial DNA. While species-specific qPCR enables precise detection, future studies should consider incorporating broad-range bacterial quantification to allow comparison of target species relative to the overall microbiome load. This would enhance interpretability and enable better normalization across samples and studies. In addition, we observed a statistically significant difference in educational level between MS patients and controls, which may reflect broader socioeconomic differences that could influence gut microbiota. Although our analyses adjusted for this variable, we acknowledge it as a potential confounder requiring further exploration in future studies. Although we used qPCR-based quantification to approximate CFU-equivalents, this molecular method detects total bacterial DNA and does not distinguish between live and dead cells, which is a recognized limitation in microbiome studies employing non-culture-based approaches.

## Future directions

Future research should focus on longitudinal studies to determine whether gut dysbiosis precedes MS onset or develops as a result of disease progression. Incorporating dietary assessments and controlling for environmental factors in larger, more diverse cohorts could strengthen the reliability of findings. Mechanistic studies exploring microbial metabolites, such as SCFAs and polysaccharide A, are also warranted to better understand how gut microbiota influence neuroinflammation and immune regulation. Additionally, interventions targeting the gut microbiota, such as probiotics or dietary modifications, could be explored as potential therapeutic strategies for MS.

## Conclusion

In conclusion, these results suggest a potential link between gut dysbiosis and MS pathogenesis. However, the study establishes correlation, not causation. The findings contribute to the growing evidence on the gut-brain axis in MS and may lead to new therapeutic approaches. Further research, including larger longitudinal studies, is needed to clarify the relationship between gut microbiota and MS.

## Data availability

All the data generated and analyzed during the current study are not publicly available but are available from the corresponding author upon reasonable request.

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## Author contributions

PR designed and led the study, personally funded the laboratory facilities, and supervised the research process. She also performed the statistical analysis and contributed to manuscript writing. CA and HH assisted with sample collection and initial data organization and manuscript writing. MZ and HS contributed to study design and manuscript drafting. KhD assisted with statistical validation and provided insights into the study's clinical implications. BG and NMK served as corresponding authors, coordinated the overall manuscript development, and handled communications during submission and review.

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## Declarations

## Competing interests

The authors declare no competing interests.

## Additional information

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